Keven Patrick Almeida Garcia

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LinkedIn GitHub Portfolio

About Me

I am currently pursuing a Bachelor's degree in Network and Computer Systems Engineering. My main interests lie in IoT, embedded systems, cybersecurity, penetration testing, mobile and web development, and cyber-physical systems. I am passionate about contributing to innovative and challenging projects that solve real-world problems. With a strong focus on developing practical solutions, I am eager to apply my skills in technology to drive progress and create impactful systems.

Education

Bachelor's in Computer Network and Systems Engineering **07/2022 – Present**
Polytechnic Institute of Viana do Castelo *Viana do Castelo, Portugal*

Certificates

- CCNA: Switching, Routing, and Wireless Essentials
- CCNA: Enterprise Networking, Security, and Automation

Language Skills

- Mother tongue(s): Criolo, Português
- Other Languages: English (B1 - Intermediate)

Skills

- Programming Languages: C,Java, Kotlin
- Web Development: HTML, CSS, JavaScript, Node.js, Express.js, Bootstrap
- Database Management: MySQL, PostgreSQL
- Embedded Systems: Arduino, Microcontrollers (ESP32)
- Cyber-Physical Systems: IoT Device Integration, Sensor Networks, Actuator Control
- Communication Protocols: MQTT, CoAP, HTTP/REST, WebSocket, TCP/IP
- Networking & Security: Cisco Networking, Virtualization,Pentesting

Projects

My Ticket - Event Ticket Sales System **04/2024 – 06/2024**
HTML, CSS, JavaScript, Node.js, Express, Prisma, PostgreSQL
- Developed a responsive platform for event ticket sales, with PostgreSQL database integration.
- Implemented CRUD functionality and APIs for managing tickets, events, and customers.
- Used Node.js, Express, and Prisma for backend development and database integration.
- Technologies: HTML, CSS, JavaScript, Node.js, Express, Prisma, PostgreSQL.

Irrigation System **12/2024 – 01/2025**
ESP32, Node.js, Express, HTML, CSS, JavaScript, Arduino C
- Developed an intelligent irrigation system using an ESP32 microcontroller and a humidity sensor to monitor soil moisture.
- Built a backend server with Node.js and Express to receive sensor data and display it in real-time on the dashboard.
- Created Arduino C code using WiFi.h and HTTPClient.h libraries for network communication.
- Technologies: ESP32, Node.js, Express, Arduino C, HTML, CSS, JavaScript.

Internship Opportunities App 11/2024 – 01/2025

Kotlin, Android, Coroutines, Retrofit, Google Maps API, Room, LiveData, RecyclerView

- Developed an Android application using Kotlin that displays companies offering internships to IPVC students, powered by an API with Retrofit.
- Implemented Coroutines for asynchronous programming and used Retrofit to fetch data from the API.
- Integrated Google Maps API for displaying company locations and utilized RecyclerView for dynamic listing of internships.
- Used Room for local database storage and LiveData to update UI in real-time based on data changes.
- Technologies: Kotlin, Android, Coroutines, Retrofit, Google Maps API, Room, LiveData, RecyclerView.

Real-Time Sensor Communication Platform 10/2025 - 01/2026

C/C++, Teensy 4.1, NativeEthernet, Node.js, Express, WebSocket, TCP/IP, HTML, CSS, JavaScript

- Developed an end-to-end system for real-time acquisition, transmission, and visualization of sensor data using a Teensy 4.1 microcontroller and a Node.js gateway.
- Implemented a custom TCP protocol with fixed-size frames (32 bytes), including header, payload, CRC16 validation, and tail for reliable data transmission.
- Designed firmware with interrupt-based data acquisition, ring buffer, and double buffering to ensure continuous and stable data streaming.
- Built a Node.js application to decode frames, validate integrity (CRC16), detect packet loss/duplication, and compute performance metrics (SPS, throughput).
- Developed a Web interface with WebSocket communication for real-time monitoring and generation of performance reports.
- Technologies: C/C++, Teensy 4.1, NativeEthernet, Node.js, Express, WebSocket, TCP/IP, HTML, CSS, JavaScript.

Communication Simulation Between Devices 12/2023 – 01/2024

Java, Data Structures, Object-Oriented Programming

- Developed a simulation system in Java that models communication between terminal devices, such as computers and switches, within a network.
- Implemented functionality for inserting devices, configuring network connections, simulating data transmission, and displaying ARP tables.
- Used object-oriented programming principles to represent devices, switches, and network configuration.
- Technologies: Java, Data Structures, Object-Oriented Programming.

RGB LED Master-Slave Humidity Control 12/2023 – 01/2024

Arduino, I2C Protocol, RGB LED, Soil Moisture Sensor

- Developed a Master-Slave system for controlling an RGB LED based on soil humidity levels using the I2C communication protocol.
- The Master (microcontroller) processes humidity data and adjusts the LED color (blue, green, or red) based on predefined thresholds.
- Slave A controls the RGB LED, adjusting its color intensity based on the received data, while Slave B (the sensor) reads the soil moisture and sends the data to the Master.
- Technologies: Arduino, I2C Protocol, RGB LED, Soil Moisture Sensor.

Additional Skills

- Strong problem-solving abilities.
- Adaptability and quick learning.
- Attention to detail.
- Ability to work under pressure.